



Exploratory Data Analysis (EDA) Concepts

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Recap!

- Data Science!!
- R Projects/Quarto/Git/GitHub for reproducibility/communication
- R Data Structures
 - Vectors, Matrices, Data Frames, Lists
- R Control Flow
 - if/then/else, loops, function writing
- Reading & Manipulating data with the tidyverse!
- Next: Gain meaningful insights from data through EDA
- Later: Dashboards, Predictive Modeling, & More

EDA Basics

- Get to know your data!
- EDA generally consists of a few steps:
 - Understand how your data is stored
 - Do basic data validation
 - Determine rate of missing values
 - Clean data up data as needed
 - Investigate distributions
 - Univariate measures/graphs
 - Multivariate measures/graphs
 - Apply transformations and repeat previous step

Understand How Data is Stored

Let's read in some data!

- **Appendicitis Data**

This dataset was acquired in a retrospective study from a cohort of pediatric patients admitted with abdominal pain to Children's Hospital St. Hedwig in Regensburg, Germany. ... Alongside multiple US images for each subject, the dataset includes information encompassing laboratory tests, physical examination results, clinical scores, such as Alvarado and pediatric appendicitis scores, and expert-produced ultrasonographic findings. Lastly, the subjects were labeled w.r.t. three target variables: diagnosis (appendicitis vs. no appendicitis), management (surgical vs. conservative) and severity (complicated vs. uncomplicated or no appendicitis). ...

Understand How Data is Stored

```
#download data to local folder
library(tidyverse)
library(readxl)
app_data <- read_excel("data/app_data.xlsx", sheet = 1)
```

- Column data types should make sense for what you expect!

app_data

```
## # A tibble: 782 x 58
##   Age BMI Sex Height Weight Length_of_Stay Management Severity
##   <dbl> <chr> <chr> <dbl> <dbl> <dbl> <chr> <chr>
## 1 12.7 16.899999999999999~ fema~ 148 37 3 conservat~ uncompl~
## 2 14.1 31.9 male 147 69.5 2 conservat~ uncompl~
## 3 14.1 23.3 fema~ 163 62 4 conservat~ uncompl~
## 4 16.4 20.6 fema~ 165 56 3 conservat~ uncompl~
## 5 11.1 16.899999999999999~ fema~ 163 45 3 conservat~ uncompl~
## # i 777 more rows
## # i 50 more variables: Diagnosis_Presumptive <chr>, Diagnosis <chr>,
## # Alvarado_Score <dbl>, Paedriatic_Appendicitis_Score <dbl>,
## # Appendix_on_US <chr>, Appendix_Diameter <dbl>, Migratory_Pain <chr>,
## # Lower_Right_Abd_Pain <chr>, Contralateral_Rebound_Tenderness <chr>,
## # Coughing_Pain <chr>, Nausea <chr>, Loss_of_Appetite <chr>,
## # C_Count <dbl>, Neutrophil_Percentage <dbl>, ...
```

Understand How Data is Stored

- Check the structure of the data!

```
str(app_data)
```

```
## tibble [782 x 58] (S3: tbl_df/tbl/data.frame)
##  $ Age                : num [1:782] 12.7 14.1 14.1 16.4 11.1 ...
##  $ BMI                 : chr [1:782] "16.899999999999999" "31.9" "23.3" "20.6" ...
##  $ Sex                  : chr [1:782] "female" "male" "female" "female" ...
##  $ Height               : num [1:782] 148 147 163 165 163 121 140 NA 131 174 ...
##  $ Weight               : num [1:782] 37 69.5 62 56 45 45 38.5 21.5 26.7 45.5 ...
##  $ Length_of_Stay       : num [1:782] 3 2 4 3 3 3 3 2 3 3 ...
##  $ Management           : chr [1:782] "conservative" "conservative" "conservative" "conservative" ...
##  $ Severity             : chr [1:782] "uncomplicated" "uncomplicated" "uncomplicated" "uncomplicated" ...
##  $ Diagnosis_Presumptive : chr [1:782] "appendicitis" "appendicitis" "appendicitis" "appendicitis" ...
##  $ Diagnosis            : chr [1:782] "appendicitis" "no appendicitis" "no appendicitis" "no appendicitis" ..
##  $ Alvarado_Score        : num [1:782] 4 5 5 7 5 6 5 3 7 4 ...
##  $ Paedriatic_Appendicitis_Score : num [1:782] 3 4 3 6 6 7 6 3 6 4 ...
##  $ Appendix_on_US        : chr [1:782] "yes" "no" "no" "no" ...
##  $ Appendix_Diameter     : num [1:782] 7.1 NA NA NA 7 NA NA NA 3.7 8 ...
##  $ Migratory_Pain        : chr [1:782] "no" "yes" "no" "yes" ...
##  $ Lower_Right_Abd_Pain  : chr [1:782] "yes" "yes" "yes" "yes" ...
##  $ Contralateral_Rebound_Tenderness : chr [1:782] "yes" "yes" "yes" "no" ...
##  $ Coughing_Pain         : chr [1:782] "no" "no" "no" "no" ...
##  $ Nausea                : chr [1:782] "no" "no" "no" "yes" ...
##  $ Vomiting              : chr [1:782] "yes" "yes" "no" "yes" ...
##  $ Neutrophil_Percentage : num [1:782] 37 36.9 36.6 36 36.9 36.9 36.7 36.8 37.3 37.1 ...
##  $ C-Reactive_Protein     : num [1:782] 7.7 8.1 13.2 11.4 8.1 9.5 10 8 20.9 5.8 ...
##  $ Hemoglobin              : num [1:782] 68.2 64.8 74.8 63 44 71.4 69.1 79.6 76 47.2 ...
##  $ Hematocrit              : num [1:782] NA NA NA NA NA NA NA NA NA NA ...
##  $ Hemoglobin_1            : num [1:782] NA NA NA NA NA NA NA NA NA NA ...
```

Convert Columns Explicitly

- `as.*()` family of functions can help coerce columns to the correct type

```
app_data <- app_data |>
  mutate(BMI = as.numeric(BMI),
         US_Number = as.character(US_Number))
app_data
```

```
## # A tibble: 782 x 58
##   Age   BMI Sex   Height Weight Length_of_Stay Management   Severity
##   <dbl> <dbl> <chr>   <dbl> <dbl>      <dbl> <chr>      <chr>
## 1  12.7  16.9 female   148    37          3 conservative uncomplicated
## 2  14.1  31.9 male     147   69.5        2 conservative uncomplicated
## 3  14.1  23.3 female   163    62          4 conservative uncomplicated
## 4  16.4  20.6 female   165    56          3 conservative uncomplicated
## 5  11.1  16.9 female   163    45          3 conservative uncomplicated
## # i 777 more rows
## # i 50 more variables: Diagnosis_Presumptive <chr>, Diagnosis <chr>,
## #   Alvarado_Score <dbl>, Paedriatic_Appendicitis_Score <dbl>,
## #   Appendix_on_US <chr>, Appendix_Diameter <dbl>, Migratory_Pain <chr>,
## #   Lower_Right_Abd_Pain <chr>, Contralateral_Rebound_Tenderness <chr>,
## #   Coughing_Pain <chr>, Nausea <chr>, Loss_of_Appetite <chr>,
## #   Body_Temperature <dbl>, WBC_Count <dbl>, Neutrophil_Percentage <dbl>, ...
```

Do Basic Data Validation

- Can use the `psych::describe()` function
- Check that the min's, max's, etc. all make sense!

```
psych::describe(app_data)
```

	vars	n	mean	sd	median	trimmed	mad
## Age	1	781	11.35	3.53	11.44	11.53	3.59
## BMI	2	755	18.91	4.39	18.06	18.43	3.91
## Sex*	3	780	1.52	0.50	2.00	1.52	0.00
## Height	4	756	148.02	19.73	149.65	149.33	19.50
## Weight	5	779	43.17	17.39	41.40	42.18	18.68
## Length_of_Stay	6	778	4.28	2.57	3.00	3.85	1.48
## Management*	7	781	1.42	0.57	1.00	1.35	0.00
## Severity*	8	781	1.85	0.36	2.00	1.93	0.00
## Diagnosis_Presumptive*	9	780	4.04	2.86	3.00	3.17	0.00
## Diagnosis*	10	780	1.41	0.49	1.00	1.38	0.00
## Alvarado_Score	11	730	5.92	2.16	6.00	5.96	2.97
## Paedriatic_Appendicitis_Score	12	730	5.25	1.96	5.00	5.21	1.48
## Appendix_on_US*	13	777	1.65	0.48	2.00	1.69	0.00
## Appendix_Diameter	14	498	7.76	2.54	7.50	7.63	2.22
## Migratory_Pain*	15	773	1.27	0.45	1.00	1.22	0.00
## Lower_Right_Abd_Pain*	16	774	1.95	0.22	2.00	2.00	0.00
## Contralateral_Rebound_Tenderness*	17	767	1.39	0.49	1.00	1.36	0.00
## Coughing_Pain*	18	766	1.28	0.45	1.00	1.23	0.00
## WBC_Count	19	774	1.59	0.49	2.00	1.61	0.00
	20	772	1.51	0.50	2.00	1.51	0.00
	21	775	37.40	0.90	37.20	37.36	0.74
	22	776	12.67	5.37	12.00	12.26	5.78

Determine Rate of Missing Values

- Use `is.na()`

```
colSums(is.na(app_data))
```

```
##           Age           BMI
##           1           27
##           Sex           Height
##           2           26
##           Weight       Length_of_Stay
##           3           4
##           Management   Severity
##           1           1
##           Diagnosis_Presumptive Diagnosis
##           2           2
##           Alvarado_Score Paedriatic_Appendicitis_Score
##           52           52
##           Appendix_on_US Appendix_Diameter
##           5           284
##           Migratory_Pain Lower_Right_Abd_Pain
##           9           8
##           Contralateral_Rebound_Tenderness Coughing_Pain
##           15          16
##           Nausea       Loss_of_Appetite
##           8           10
##           re           WBC_Count
##           7           6
##           ge           Segmented_Neutrophils
##           103          728
```

Determine Rate of Missing Values

- Stay in the tidyverse

```
sum_na <- function(column){  
  sum(is.na(column))  
}  
na_counts <- app_data |>  
  summarize(across(everything(), sum_na))  
na_counts
```

```
## # A tibble: 1 x 58  
##   Age BMI Sex Height Weight Length_of_Stay Management Severity  
##   <int> <int> <int> <int> <int> <int> <int> <int>  
## 1 1 27 2 26 3 4 1 1  
## # i 50 more variables: Diagnosis_Presumptive <int>, Diagnosis <int>,  
## # Alvarado_Score <int>, Paedriatic_Appendicitis_Score <int>,  
## # Appendix_on_US <int>, Appendix_Diameter <int>, Migratory_Pain <int>,  
## # Lower_Right_Abd_Pain <int>, Contralateral_Rebound_Tenderness <int>,  
## # Coughing_Pain <int>, Nausea <int>, Loss_of_Appetite <int>,  
## # Body_Temperature <int>, WBC_Count <int>, Neutrophil_Percentage <int>,  
## # Segmented_Neutrophils <int>, Neutrophilia <int>, RBC_Count <int>, ...
```

Clean Up Data As Needed

- Can remove rows with missing using `tidyr::drop_na()` function

```
names(app_data)[na_counts < 30]
```

```
## [1] "Age"
## [3] "Sex"
## [5] "Weight"
## [7] "Management"
## [9] "Diagnosis_Presumptive"
## [11] "Appendix_on_US"
## [13] "Lower_Right_Abd_Pain"
## [15] "Coughing_Pain"
## [17] "Loss_of_Appetite"
## [19] "WBC_Count"
## [21] "Hemoglobin"
## [23] "Thrombocyte_Count"
## [25] "Dysuria"
## [27] "Peritonitis"
## [29] "US_Number"
## [1] "BMI"
## [3] "Height"
## [5] "Length_of_Stay"
## [7] "Severity"
## [9] "Diagnosis"
## [11] "Migratory_Pain"
## [13] "Contralateral_Rebound_Tenderness"
## [15] "Nausea"
## [17] "Body_Temperature"
## [19] "RBC_Count"
## [21] "RDW"
## [23] "CRP"
## [25] "Stool"
## [27] "US_Performed"
```

Clean Up Data As Needed

- Can remove rows with missing using `tidyr::drop_na()` function

```
app_data |>
  drop_na(names(app_data)[na_counts < 30])
```

```
## # A tibble: 674 x 58
##   Age   BMI Sex   Height Weight Length_of_Stay Management   Severity
##   <dbl> <dbl> <chr>   <dbl>   <dbl>         <dbl> <chr>         <chr>
## 1  12.7  16.9 female    148     37             3 conservative uncomplicated
## 2  14.1  31.9 male      147    69.5            2 conservative uncomplicated
## 3  14.1  23.3 female    163     62             4 conservative uncomplicated
## 4  16.4  20.6 female    165     56             3 conservative uncomplicated
## 5  11.1  16.9 female    163     45             3 conservative uncomplicated
## # i 669 more rows
## # i 50 more variables: Diagnosis_Presumptive <chr>, Diagnosis <chr>,
## #   Alvarado_Score <dbl>, Paedriatic_Appendicitis_Score <dbl>,
## #   Appendix_on_US <chr>, Appendix_Diameter <dbl>, Migratory_Pain <chr>,
## #   Lower_Right_Abd_Pain <chr>, Contralateral_Rebound_Tenderness <chr>,
## #   Coughing_Pain <chr>, Nausea <chr>, Loss_of_Appetite <chr>,
## #   Body_Temperature <dbl>, WBC_Count <dbl>, Neutrophil_Percentage <dbl>, ...
```

May Want to Impute Values

- We lose information when removing rows!
- Can **impute** missing values with `tidyr::replace_na()`

```
app_data <- app_data |>
  replace_na(list(BMI = mean(app_data$BMI, na.rm = TRUE),
                        Height = mean(app_data$Height, na.rm = TRUE)))
app_data
```

```
## # A tibble: 782 x 58  
##   Age BMI Sex Height Weight Length_of_Stay Management Severity  
##   <dbl> <dbl> <chr>    <dbl>    <dbl>         <dbl> <chr>      <chr>  
## 1 12.7 16.9 female     148     37             3 conservative uncomplicated  
## 2 14.1 31.9 male       147    69.5           2 conservative uncomplicated  
## 3 14.1 23.3 female     163     62           4 conservative uncomplicated  
## 4 16.4 20.6 female     165     56           3 conservative uncomplicated  
## 5 11.1 16.9 female     163     45           3 conservative uncomplicated  
## # i 777 more rows  
## # i 50 more variables: Diagnosis_Presumptive <chr>, Diagnosis <chr>,  
## # Alvarado_Score <dbl>, Paedriatic_Appendicitis_Score <dbl>,  
## # Appendix_on_US <chr>, Appendix_Diameter <dbl>, Migratory_Pain <chr>,  
## # Lower_Right_Abd_Pain <chr>, Contralateral_Rebound_Tenderness <chr>,  
## # Coughing_Pain <chr>, Nausea <chr>, Loss_of_Appetite <chr>,  
## # ...
```

EDA Basics

- Get to know your data!
- EDA generally consists of a few steps:
 - Understand how your data is stored
 - Do basic data validation
 - Determine rate of missing values
 - Clean data up data as needed
 - Investigate distributions
 - Univariate measures/graphs
 - Multivariate measures/graphs
 - Apply transformations and repeat previous step

Investigate distributions

- How to summarize data depends on the type of data
 - Categorical (Qualitative) variable - entries are a label or attribute
 - Numeric (Quantitative) variable - entries are a numerical value where math can be performed

Investigate distributions

- How to summarize data depends on the type of data
 - Categorical (Qualitative) variable - entries are a label or attribute
 - Numeric (Quantitative) variable - entries are a numerical value where math can be performed
- Numerical summaries (across subgroups)
 - Contingency Tables (for categorical data)
 - Mean/Median
 - Standard Deviation/Variance/IQR
 - Quantiles/Percentiles
- Graphical summaries (across subgroups)
 - Bar plots (for categorical data)
 - Histograms
 - Box plots
 - Scatter plots

Categorical Data

Goal: Describe the **distribution** of the variable

- Distribution = pattern and frequency with which you observe a variable
- Categorical variable - entries are a label or attribute
 - Describe the relative frequency (or count) for each category

Variables of interest for this section:

- Sex, Diagnosis, Severity

Factors

A factor variable is really useful for certain categorical variables!

Factor - special class of vector with a `levels` attribute

- Can have more descriptive labels, ordering of categories, etc.
- Levels define **all** possible values for that variable
 - Great for variable like `Day` (Monday, Tuesday, ..., Sunday)
 - Not great for variable like `Name` where new values may come up
- Great for plotting as you can order the levels and give nicer labels

Factors

- Let's create factor versions of our three variables

```
unique(app_data$Sex)
```

```
## [1] "female" "male"   NA
```

```
unique(app_data$Diagnosis)
```

```
## [1] "appendicitis" "no appendicitis" NA
```

```
unique(app_data$Severity)
```

```
## [1] "uncomplicated" NA "complicated"
```

- Now we can use `factor()` or `as.factor()` to coerce the character variables

Factors

- Let's create factor versions of our three variables

```
app_data |>
  mutate(SexF = factor(Sex, levels = c("female", "male"), labels = c("Female", "Male")),
         DiagnosisF = as.factor(Diagnosis),
         SeverityF = as.factor(Severity)) |>
  select(SexF, DiagnosisF, SeverityF)
```

```
## # A tibble: 782 x 3
##   SexF      DiagnosisF      SeverityF
##   <fct>    <fct>         <fct>
## 1 Female appendicitis uncomplicated
## 2 Male   no appendicitis uncomplicated
## 3 Female no appendicitis uncomplicated
## 4 Female no appendicitis uncomplicated
## 5 Female appendicitis  uncomplicated
## # i 777 more rows
```

Contingency Tables

- Summarize categorical data by looking at counts!

```
app_data |>
  group_by(SexF) |>
  drop_na(SexF) |>
  summarize(count = n())
```

```
## # A tibble: 2 x 2
##   SexF    count
##   <fct> <int>
## 1 Female   377
## 2 Male    403
```

```
app_data |>
  group_by(DiagnosisF) |>
  drop_na(DiagnosisF) |>
  summarize(count = n())
```

```
## # A tibble: 2 x 2
##   DiagnosisF    count
##   <fct>        <int>
## 1 appendicitis   463
## 2 no appendicitis 317
```

Contingency Tables

- Summarize categorical data by looking at counts across combinations of variables!

```
app_data |>
  group_by(SexF, DiagnosisF) |>
  drop_na(SexF, DiagnosisF) |>
  summarize(count = n()) |>
  pivot_wider(names_from = DiagnosisF, values_from = count)
```

```
## # A tibble: 2 x 3
## # Groups:   SexF [2]
##   SexF      appendicitis `no appendicitis`
##   <fct>         <int>         <int>
## 1 Female           200           176
## 2 Male             262           141
```

Contingency Tables

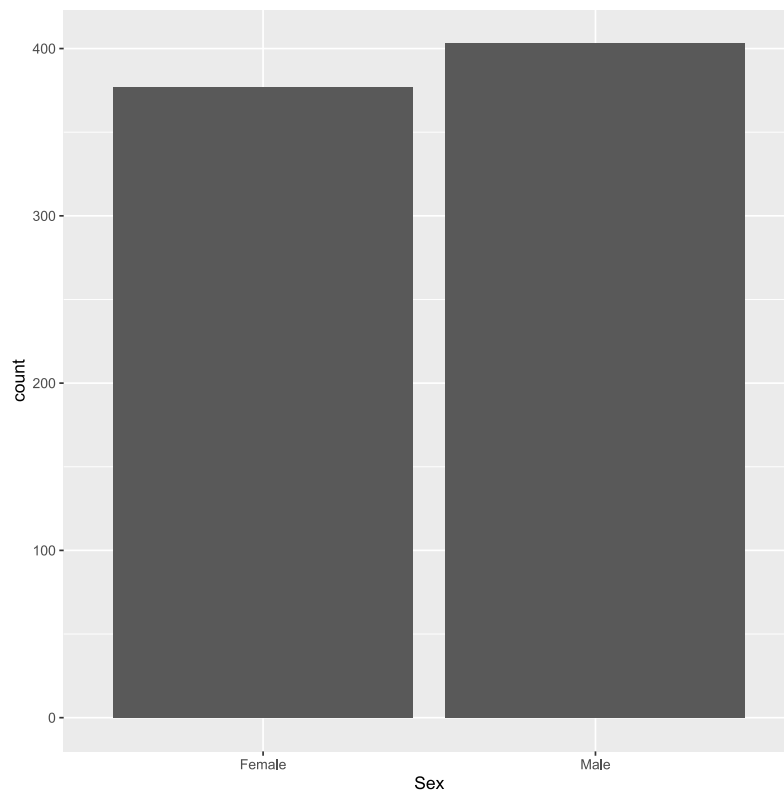
- Summarize categorical data by looking at counts across combinations of variables!

```
app_data |>
  group_by(SexF, DiagnosisF, SeverityF) |>
  drop_na(SexF, DiagnosisF, SeverityF) |>
  summarize(count = n()) |>
  pivot_wider(names_from = DiagnosisF, values_from = count)
```

```
## # A tibble: 4 x 4
## # Groups:   SexF [2]
##   SexF   SeverityF      appendicitis `no appendicitis`
##   <fct> <fct>          <int>          <int>
## 1 Female complicated         55             1
## 2 Female uncomplicated      145            175
## 3 Male   complicated         63             NA
## 4 Male   uncomplicated      199            141
```

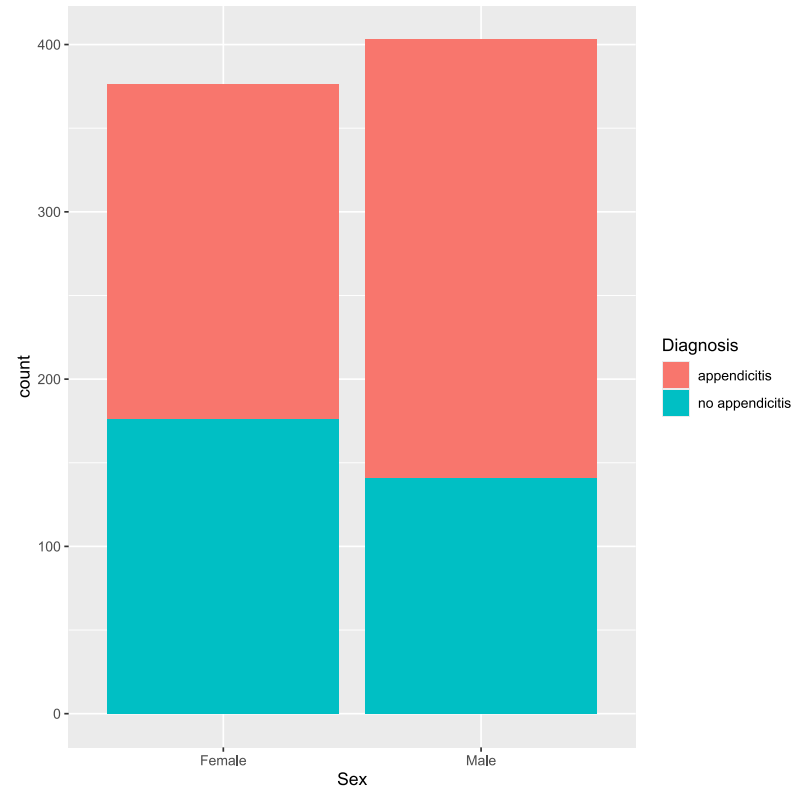
Bar Charts

- Main visual used is a bar plot! Simply displays our counts with bars.



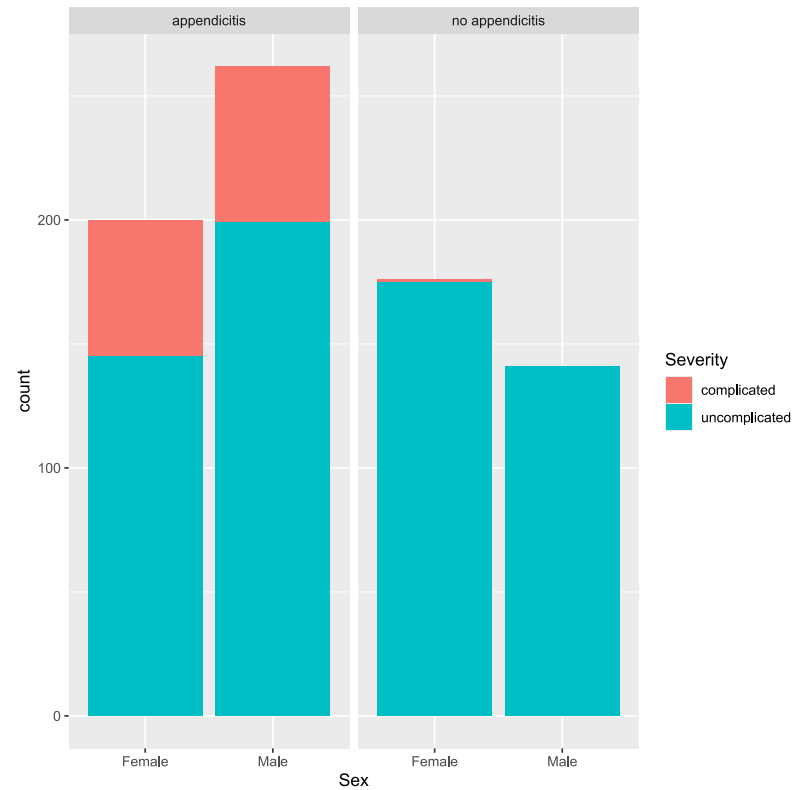
Bar Charts

- Main visual used is a bar plot! Simply displays our counts with bars.



Bar Charts

- Main visual used is a bar plot! Simply displays our counts with bars.



Numeric Data

Goal: Describe the **distribution** of the variable

- Distribution = pattern and frequency with which you observe a variable
- Numeric variable - entries are a numerical value where math can be performed

For a single numeric variable, describe the distribution via

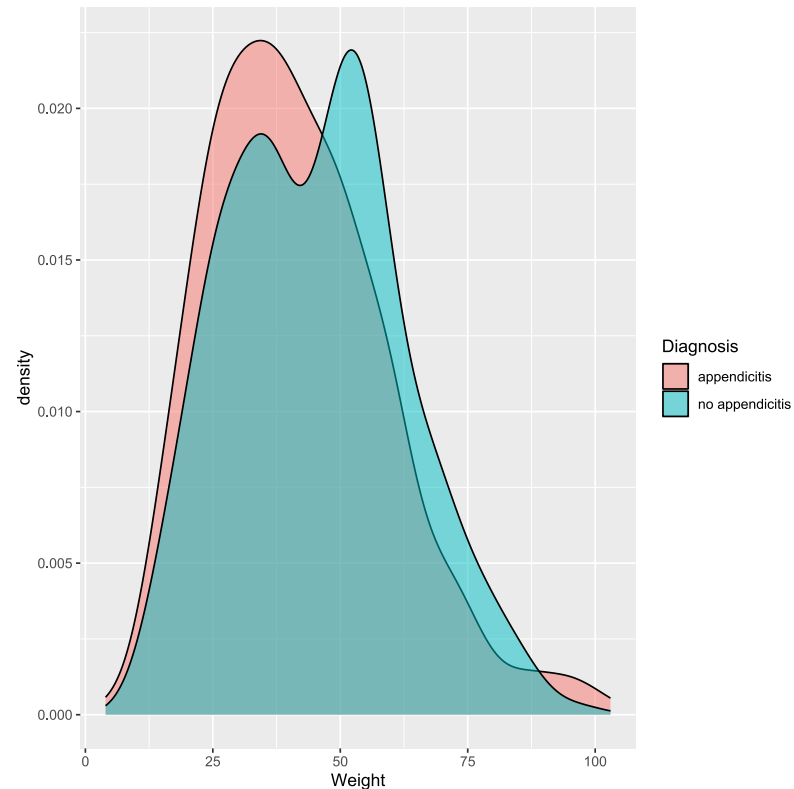
- Shape: Histogram, Density plot, ...
- Measures of center: Mean, Median, ...
- Measures of spread: Variance, Standard Deviation, Quartiles, IQR, ...

For two numeric variables, describe the distribution via

- Shape: Scatter plot, ...
- Measures of linear relationship: Covariance, Correlation

Summarizing Center and Spread

- We summarize center and spread for a numeric variable because it is difficult to compare entire distributions!
 - Consider the distributions of `Weight` for those with appendicitis and those without



Summarizing Center and Spread

- Mean and Median give good measures of the 'middle' type observations

```
app_data |>
  group_by(Diagnosis) |>
  drop_na(Diagnosis, Weight) |>
  summarize(mean_weight = mean(Weight),
            median_weight = median(Weight))
```

```
## # A tibble: 2 x 3
##   Diagnosis      mean_weight median_weight
##   <chr>          <dbl>         <dbl>
## 1 appendicitis    41.7           39.5
## 2 no appendicitis 45.3           46.3
```

Summarizing Center and Spread

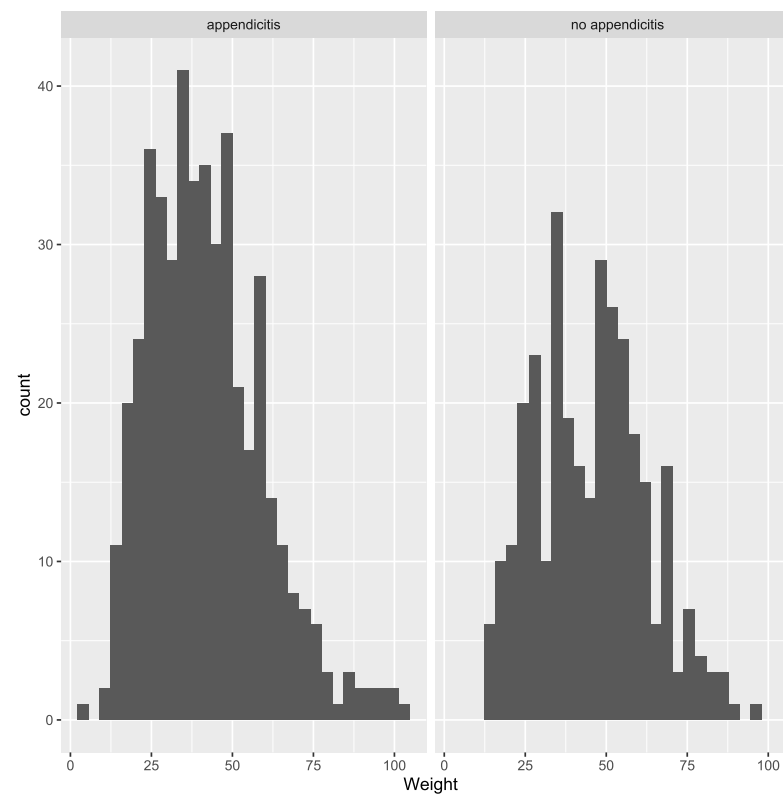
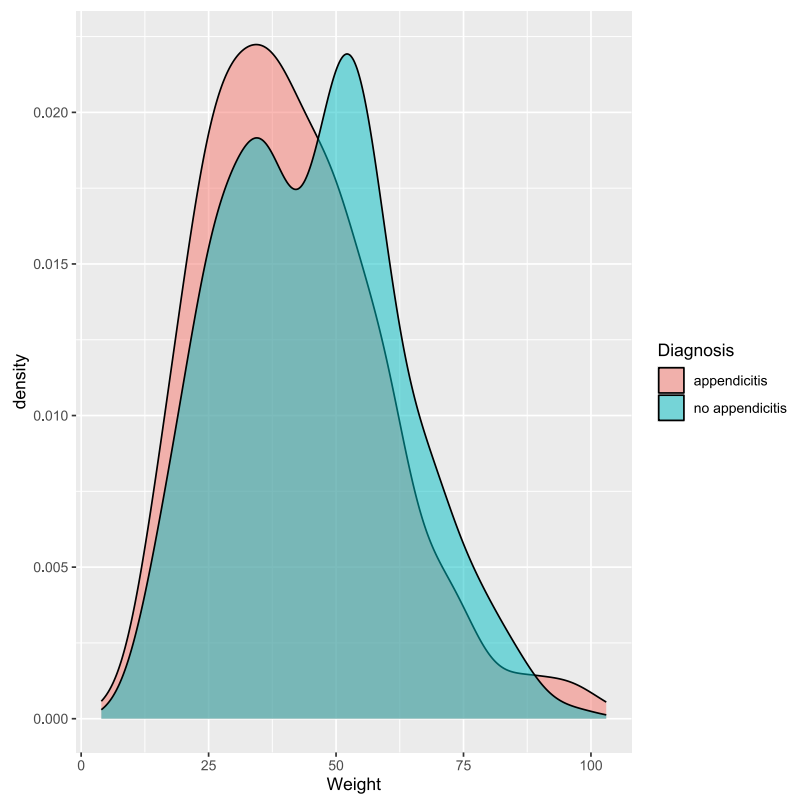
- Of course we need to understand the variability we see as well! Variance, standard deviation, and IQR are good measures of that.

```
app_data |>
  group_by(Diagnosis) |>
  drop_na(Diagnosis, Weight) |>
  summarize(across(Weight, .fns = list("mean" = mean,
                                       "median" = median,
                                       "var" = var,
                                       "sd" = sd,
                                       "IQR" = IQR), .names = "{.fn}_{.col}"))
```

```
## # A tibble: 2 x 6
##   Diagnosis      mean_Weight median_Weight var_Weight sd_Weight IQR_Weight
##   <chr>          <dbl>         <dbl>      <dbl>    <dbl>    <dbl>
## 1 appendicitis    41.7           39.5       305.     17.5     23.4
## 2 no appendicitis 45.3           46.3       293.     17.1     23.5
```

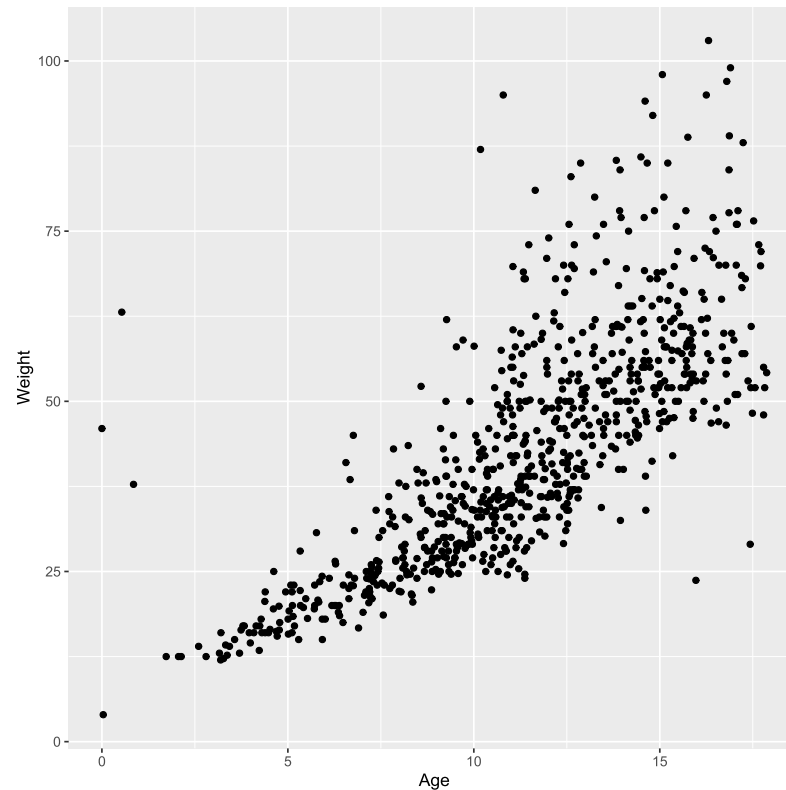
Summarizing Shape

- Most easily done via histograms and density plots
 - Histograms are more variable, which can be bad!



Summarizing Two Numeric Variables

- To look at the distribution of two numeric variables together, we usually look at a scatter plot!



Summarizing Two Numeric Variables

- Again, difficult to describe the relationship generally!
 - Numerically we commonly describe the 'linear-ness' of the relationship
 - Done through covariance and correlation

```
app_data |>
  drop_na(Weight, Age) |>
  summarize(cov = cov(Weight, Age), corr = cor(Weight, Age))
```

```
## # A tibble: 1 x 2
##   cov  corr
##   <dbl> <dbl>
## 1  47.0  0.766
```

Summarizing Two Numeric Variables

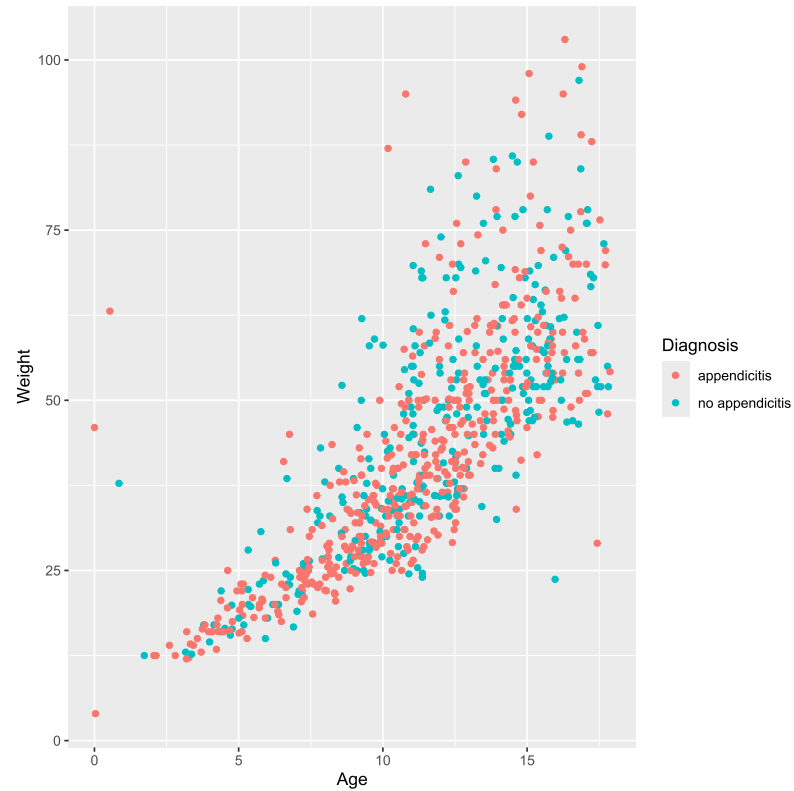
- Again, difficult to describe the relationship generally!
 - Numerically we commonly describe the 'linear-ness' of the relationship
 - Done through covariance and correlation

```
app_data |>
  drop_na(Weight, Age) |>
  summarize(cov = cov(Weight, Age), corr = cor(Weight, Age))
```

```
## # A tibble: 1 x 2
##   cov  corr
##   <dbl> <dbl>
## 1  47.0  0.766
```

Summarizing Two Numeric Variables

- Of course we want to bring in subgroups to compare them!



Summarizing Two Numeric Variables

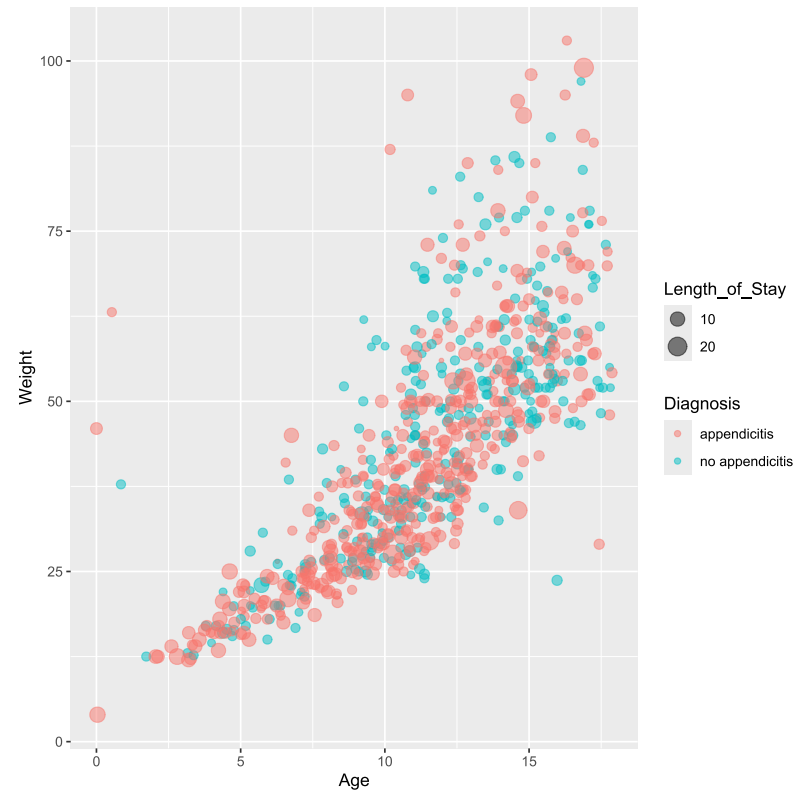
- Summarize based on groups!

```
app_data |>
  drop_na(Weight, Age, Diagnosis) |>
  group_by(Diagnosis) |>
  summarize(cov = cov(Weight, Age), corr = cor(Weight, Age))
```

```
## # A tibble: 2 x 3
##   Diagnosis      cov  corr
##   <chr>      <dbl> <dbl>
## 1 appendicitis  48.0 0.775
## 2 no appendicitis 44.3 0.748
```

Summarizing Two Numeric Variables

- We can do really interesting stuff to add in additional variables (like a third numeric variable)



Recap

- EDA is often the first step to an analysis:
 - Understand how your data is stored
 - Do basic data validation
 - Determine rate of missing values
 - Clean data up data as needed
 - Investigate distributions
 - Univariate measures/graphs
 - Multivariate measures/graphs
 - Apply transformations and repeat previous step